Application of Foundation Model and Agile Modeling to Passive Acoustic Detection of Orcas in Monterey Bay National Marine Sanctuary

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8 days
of confirmed killer-whale presence across April–May 2018

95% fewer
false alarms in a held-out month, from 17 corrective labels (6,489 → 304)

~1,450 labels
in ~8–10 h of one expert's time; the classifier retrains in ~30 s

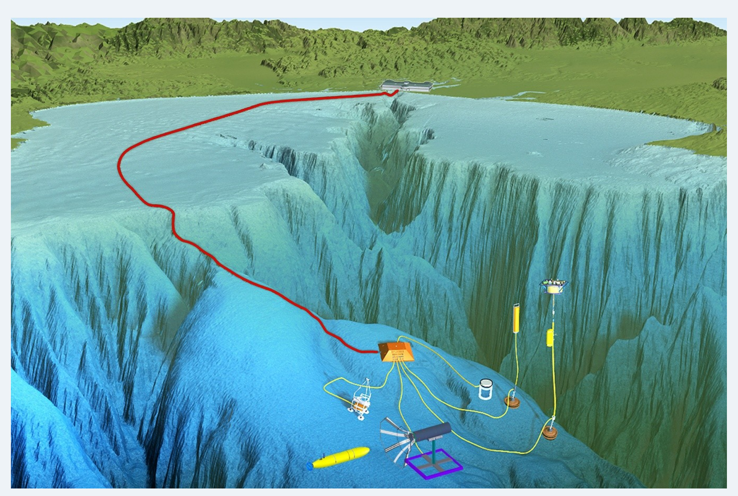
0.95
orca detection score (F1: 0 = useless, 1 = perfect) at our chosen cutoff

1 The gap

- Monterey Bay National Marine Sanctuary hosts three killer whale ecotypes: transient (Bigg's), endangered Southern Resident, and Offshore.
- None are yet routinely detected and classified from passive acoustic monitoring (PAM) data.
- Detections are rare and manual annotation of thousands of hours is impractical, so labeled MARS training data for these ecotypes are scarce.
- Goal: turn a continuous PAM archive into an ecologically useful record of orca presence — with its limits measured, not hidden.

2 A continuous listening post

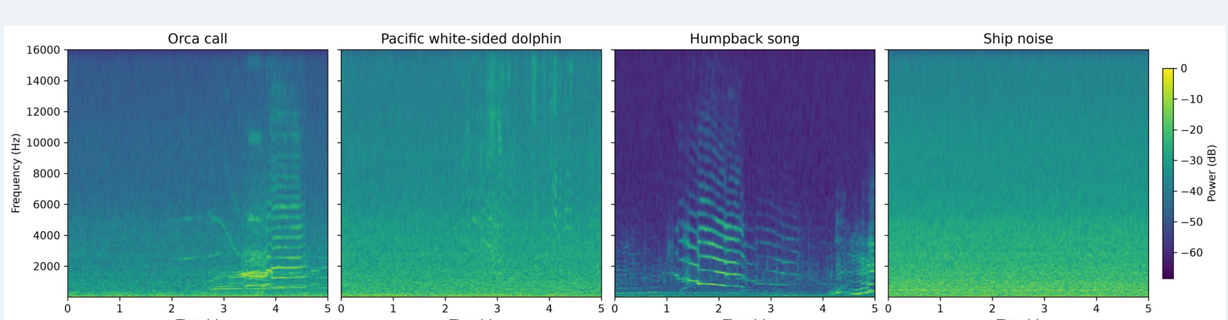
- MARS cabled observatory, Smooth Ridge (36°42.75′ N, 122°11.21′ W), ~30 km offshore at ~900 m depth.
- Ocean Sonics icListen HF omnidirectional hydrophone, 10 Hz – 100 kHz, relayed to shore in real time, 24/7.
- Archived and public as the Pacific Ocean Sound recordings; analysis windows here span 2018–2026.



The MARS cabled observatory; the hydrophone sits at the science node. Image: MBARI.

3 Off-the-shelf CNNs confuse the bay

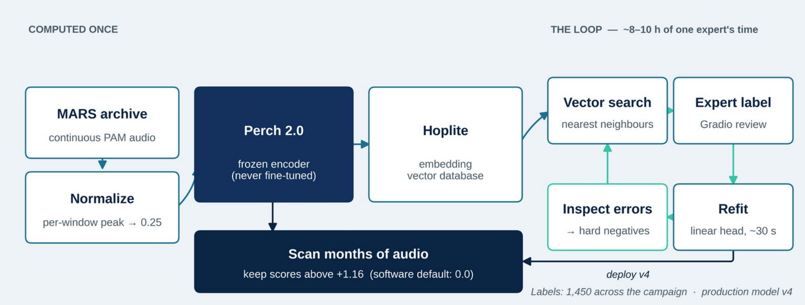
- The Google multispecies whale CNN (seven whale classes) and OrcAI (Icelandic fish-eating orcas) both produced high-confidence orca detections on MARS recordings.
- Both also confused orca calls with Pacific white-sided dolphin (Lagenorhynchus obliquidens) and other signals — confusion, not sensitivity, was the limiting factor.
- Their scores were still useful as a bootstrap, ranking candidate clips for the first round of expert labeling.



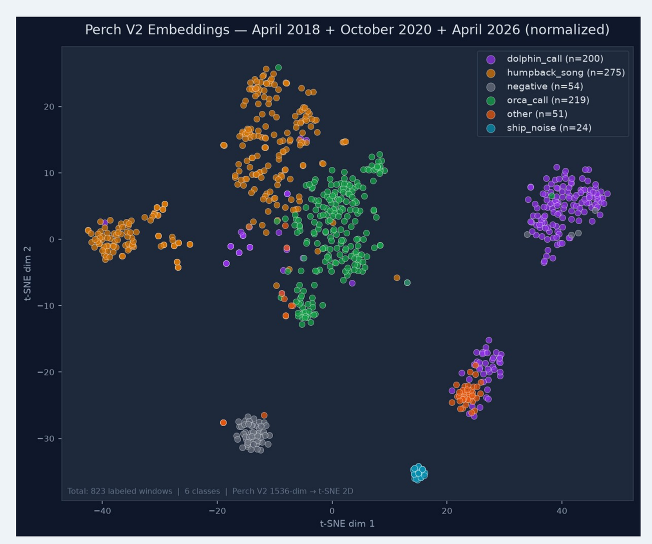
Mel-scale log-power spectrograms, one confirmed exemplar per class (5 s windows, v4 labels). All four share identical FFT parameters (2048-point FFT, 128 mel bands, 10 Hz–16 kHz) and one colour scale, so brightness and detail are directly comparable. Orca and humpback song both carry tonal, harmonically stacked content in overlapping bands — which is why the classifier confuses them.

4 Teaching a big model a new sound

- Perch 2.0 is a pre-trained bioacoustics model. It turns each 5-second window into an embedding — a 1,536-number fingerprint of the sound. Similar sounds get similar fingerprints, so one labeled example retrieves others like it from the Hoplite database.
- The loop: search → label a handful of clips → refit a simple classifier on those fixed fingerprints (~30 s) → look at its mistakes → search again. The big model is never retrained.
- Enabling fix: per-window peak normalization to 0.25 before embedding — without it, quiet distant calls were unrecoverable.
- The classifier scores every window; higher means more orca-like. We keep scores above +1.16 rather than the software's 0.0 default, with the cutoff chosen on months the model never trained on.



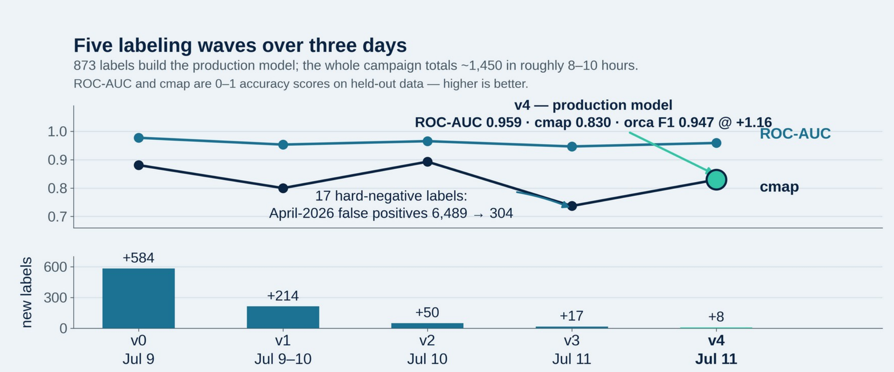
5 The classes separate



Each dot is one 5-second window, placed so that sounds the model hears as similar sit close together (t-SNE). 823 labeled windows, 6 classes, 3 seasons (April 2018, October 2020, April 2026), as labeled at the July 2026 snapshot. The orca cluster holds across all three years — the embeddings are separable across recording conditions and time, which is the premise this method rests on. Where classes visibly overlap, orca against humpback, is the same confusion the spectrograms show and the detection scores measure. The humpback class here covers vocalization broadly, not only song.

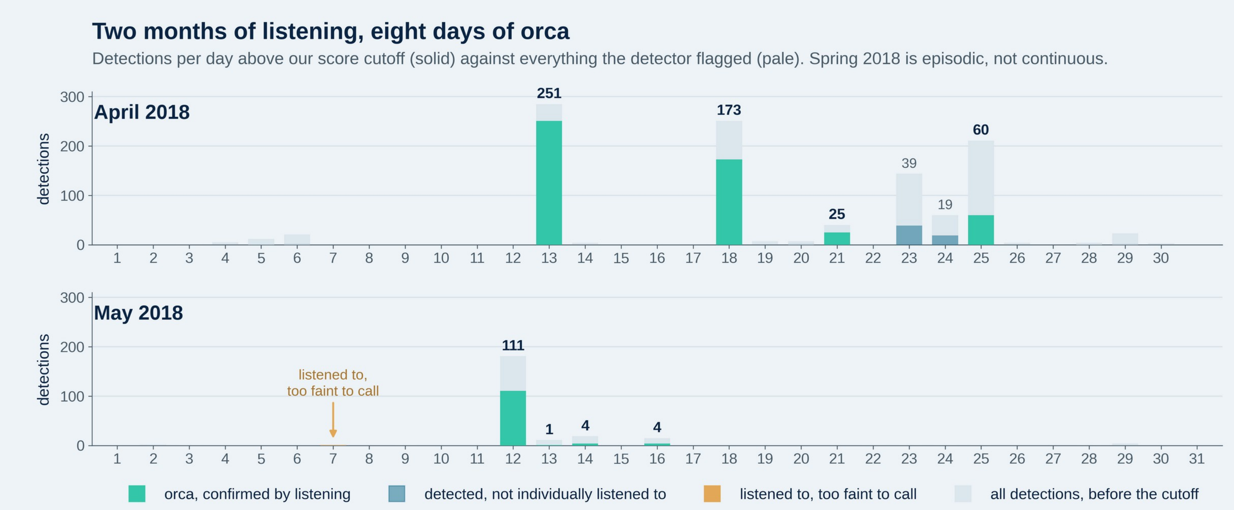
6 v0 → v4: five labeling waves

- v0 (584 labels, 5 classes) recovers the known 13 April 2018 event; v1 (+214) adds October 2020; v2 (+50) rebalances dolphin and other.
- v3 adds just 17 hard negatives — the detector's own false alarms, labeled as counter-examples — and April-2026 false alarms fall 6,489 → 304. v4 (+8) is the model used throughout this poster.
- Two further experiments degraded performance and are not presented here; detail in the project repository.



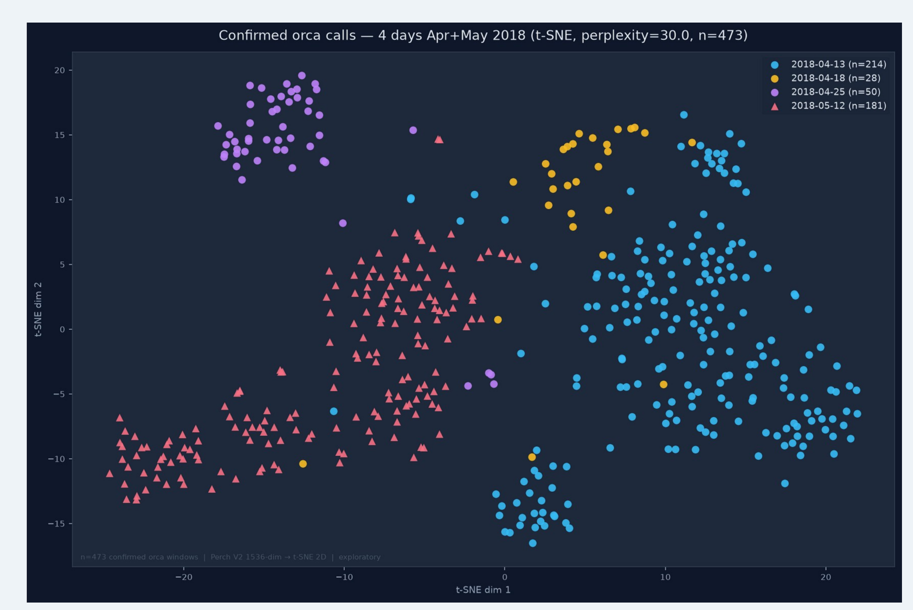
7 Spring 2018 was not one encounter

- Previously known: a single Bigg's orca event on 13 April 2018.
- Above our score cutoff, v4 also flagged 18 April (173 detections), 21 April (25) and 25 April (60). Listening to 25, 25 and 50 of those clips confirmed all 100 as orca, with no false alarms — April 2018 is a sustained two-week presence.
- May 2018: 12 May confirmed 181/181, with 13, 14 and 16 May confirmed on later review (+8 labels; 13 May rests on a single clip).
- Neither v4 nor orca_v10 has trained on May 2018 recordings — these are detections in a month the presented models have never seen.
- Eight confirmed orca days across the two months — sustained Bigg's presence, not one encounter.
- Two independent methods agree: regional news and whale-watch reports described record orca sightings that spring, and identified two pods — one Alaskan, one Californian (the CA140 matriline).



Days carry very different weight: 12 May rests on 181 reviewed clips, 13 May on one. 23 and 24 April were detected but never individually listened to, so they are not counted.

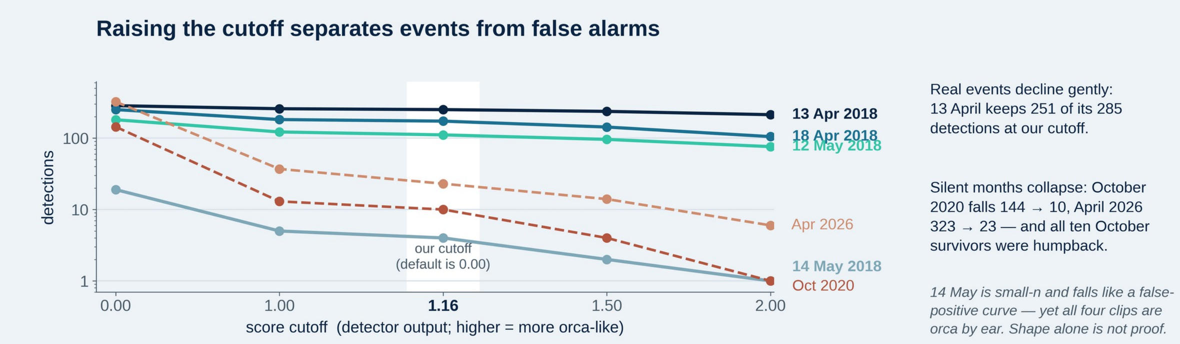
8 Structure between encounters



Zooming inside the orca cluster: 473 confirmed orca windows, coloured by day. The 25 April evening encounter (purple) sits apart from the 13 April morning event (blue), and 12 May holds its own region a month later. The classes separate in the map at left — and within orca, encounters do too. Exploratory; perplexity 30.

- The split holds however the map is drawn (three t-SNE settings) and is not an artifact of one recording or one passing vessel.
- Two pods were documented in the bay that spring; the two clusters are consistent with, not proof of, that pairing.
- The fingerprints behave like an instrument — they raise testable questions about who was calling.

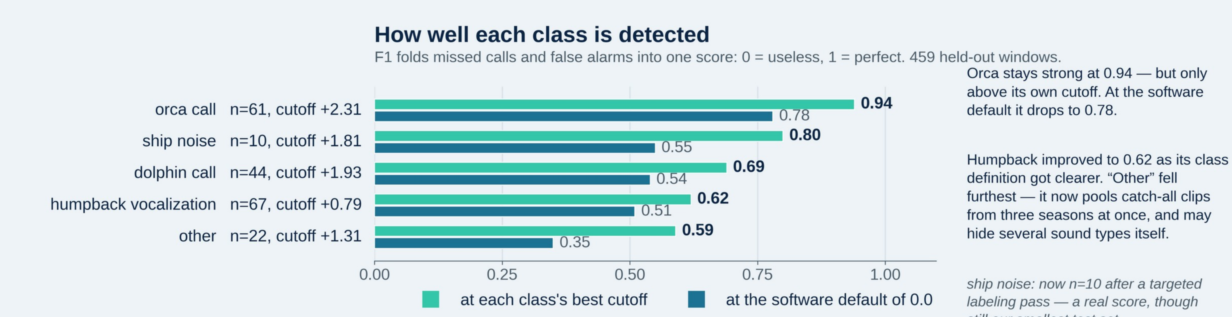
9 Seen but not heard



- Two months where orcas were seen from boats but never heard: October 2020 and April 2026.
- Every October 2020 detection above our cutoff was listened to, and every one was humpback — false alarms, not missed orca.
- Bigg's orcas are often silent while hunting, so seen-but-not-heard is a real outcome, not a broken detector. Both months now serve as a standing test that the detector stays quiet when it should.

10 Measured limits

Panels 1–9 present v4. The scores below come from orca_v10 — the same three-season recipe retrained on 1,076 labels (v4: 803), including the orca days and ship-noise pass confirmed since. Each class has its own cutoff.



- Each class needs its own cutoff — the best values run from +0.79 to +2.31, so one global setting cannot serve them all.
- Cutoffs are set on months the model did not train on, never on the month being analysed.
- Whale-watch vessels arrive once orcas are sighted, so ship noise tracks orca presence — it can never be evidence of absence.

11 Next

- Humpback vocalization is one broad class covering song and a wide range of other calls. Separating them is the likeliest route to lifting the weakest scores here.
- Then: a separate score cutoff per class, and extension to Southern Resident and Offshore ecotypes.

Acknowledgements

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Selected references

van Merriënboer et al. (2025) Perch 2.0: the bittern lesson for bioacoustics, arXiv:2508.04665 · Burns et al. (2025) Perch 2.0 transfers 'whale' to underwater tasks, arXiv:2512.03219 · Google Research, perch-hoplite (GitHub) · Harvey, Allen & Harrell (2024) multispecies-whale, Kaggle · Bonhoeffer et al. (2026) orcAI: a machine learning tool to detect and classify acoustic signals of killer whales, Mar. Mamm. Sci. 42(1) e70083, doi:10.1111/mms.70083 · Ryan et al. (2016) New passive acoustic monitoring in Monterey Bay National Marine Sanctuary, OCEANS 2016 MTS/IEEE Monterey, pp. 1–8, doi:10.1109/OCEANS.2016.7761363 · McInnes et al. (2023) Identification catalog and field guide for transient (Bigg's) killer whales of Monterey Bay and California waters, NOAA SWFSC, doi:10.25923/0q3f-1812

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QR → code, figures and data
github.com/duane-edgington/perch-hoplite

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